



JAVASCRIPT-BASED SPIROGRAPH INTERFACE

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Spirograph is a geometrical pattern generator that generates a variety of curved shapes and figures. In earlier decades, spirographs had an important application in mathematics for manual solution of polynomial equations of a higher degree. They have been used for explaining everything from trigonometric functions to gallery artworks.

There are many laser-based spirographs available in the market. Here is a simple JavaScript-based spirograph program.

How does a spirograph work

Spirograph designs are a combination of hypocycloids and epicycloids at various radii to form different patterns on paper.

Generally, spirographs available in the market consist of spire-punched worm gears, sprockets or cogwheels that have teeth on their outer peripheries which are run on a similar dented circular track. The circular worm gear acts as the rotor and the hollow circular track as the stator.

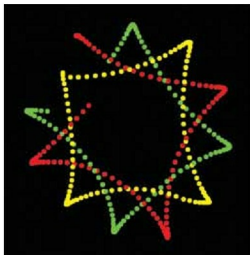
When a pencil or other marker is inserted into the punched holes on the rotor circular disk (worm gear) and then placed in the stator and moved along the circumference of the stator, at varied radii, it generates a continuous design or pattern (refer Screenshot 1).

Software program

The program depicted here uses three worm gears, namely, a, b and c, with



Screenshot 1: Spirograph



Screenshot 2: Program output spirograph

trajectory worm gear X. When the program is run and the mouse cursor moved in a circular path on the screen, a colourful spirograph pattern similar to the commercially available spirograph patterns appears on the screen. The program uses some mathematical expressions to carry out the task.

When the mouse is moved, a new coordinate is created and sent to a function called `spirograph()` that arranges the pattern according to sine and cosine functions for each ordinate and abscissa (X and Y axes) of the screen. When the mouse is paused, the pattern slowly dismantles and arranges into a moving circular wheel.

The spirograph pattern is generally the path traced by a circle's tangent as this circle rolls around the inside or along the outside of another circle at a certain speed. This motion of the circle's tangent causes epicycloids (outward) and hypocycloids (inward).

In this program, the `spirograph()` function creates epicycloids and hypocycloids. This pattern is painted on the screen by the `paint()` function which uses the main routine as its canvas to output the cycloidal spirograph patterns.

The program generates the pattern manually. That means you need to move the mouse in a circular path to generate the spirograph pattern as shown in Screenshot 2. The `track()` function traces the path of the mouse-cursor coordinate and paints the same on the screen as the cursor is moved.

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on www.efymag.com website.

The author is a B.Tech in Electronics and Communications Engineering, likes coding JavaScript and has keen interest in Space Technology (Satellite Communications) and Mobile Communication

SPIROGRAPH.HTM

```
<html>
<head>
<title>Spirograph Generator in JavaScript</title>
```

```
<meta http-equiv="Page-Enter" content="true"
evalTrans(Duration=1.0,Transition=23)">
<meta http-equiv="Page-Exit" content="true"
blendTrans(Duration=1.0)">
```

```
<meta http-equiv="Page-Exit" content="true"
evalTrans(Duration=1.0,Transition=23)">
<noscript style="font-family:Lucida Sans
Unicode; font-size: 10pt; color: Sienna
```